

Cybersecurity Incident Report

Case File: Application Layer DoS Attack

Prepared By: Maltiti Azindoo Alidu, SOC Analyst

Date: March 15, 2026

Target System: www.yummyrecipesforme.com

Status: Resolved

Executive Summary

At approximately 14:32 UTC, internal monitoring detected a critical spike in CPU utilization (99.9%) and severe network latency on the primary web server hosting www.yummyrecipesforme.com. Customers reported an inability to access the site, experiencing timeout errors. An immediate SOC investigation revealed an Application Layer (Layer 7) Denial-of-Service attack targeting the application's search functionality. The malicious traffic was identified, isolated, and blocked, restoring normal service.

Investigation Findings

Analysis of the web server access logs using Linux CLI tools (grep, awk) revealed an HTTP Flood attack originating from a single IP address (198.51.100.42).

- The attacker aggressively targeted the `/search?q=dinner` endpoint.
- The volume of resource-intensive database queries exhausted server capacity, resulting in 503 Service Unavailable errors.
- Legitimate user traffic timed out, returning 408 Request Timeout errors.

Actions Taken

1. **Detection:** Parsed access.log to isolate the top-talking IP address.
2. **Containment:** Implemented a network-level block on IP 198.51.100.42 using iptables firewall rules.
3. **Recovery:** Monitored server CPU levels until they returned to normal operational baselines (12.4%). Verified website accessibility from external networks.

Recommendations

To bolster resilience against future volumetric attacks, the security engineering team should implement web server rate limiting and deploy a Web Application Firewall (WAF) to automate the detection and blocking of abnormal HTTP request patterns.